RESPONSE TO CLAUDE | 26 OCTOBER 2026

Tensile Workbench Audit response

Accepted findings, proposed remediations and qualified recommendations

The audit identifies credible risks to specimen attribution, units and calculated results. We accept its central conclusion: passing the existing tests does not establish that these failure cases are handled correctly. Several findings reproduce independently in the current supported environment.

Status: This document records review decisions and proposed work. None of the remediations below has been implemented as part of this response. "Accepted" means the finding is accepted, not that it is fixed or that every suggested remedy is approved.

ID	Finding	Disposition
C1	Silent pooling of nested sample groups	Accept; clarify group boundaries
C2	Duplicate exports counted twice	Accept; qualify automatic exclusion
C3	Progressive failure misclassified as sudden	Accept; validate replacement criteria
C4	Fixed elastic-fit window biases YS and E	Accept limitation; qualify fitting remedy
C5	Percent strain incorrectly multiplied by 100	Accept; prioritise explicit units
C6	Stress units and true/engineering confusion	Accept; prioritise explicit interpretation
C7	Unverified Instron assignment still used	Accept; gate dependent values
C8	Misleading inferred specimen locations	Accept ambiguity; qualify plausibility rules
C9	Excluded values remain numeric in exports	Accept; formatting alone is insufficient
C10	Import failures hidden; counts misleading	Accept; separate tested and loaded counts

Review basis. Claude audited commit **fdcb232**. This response checks the subsequent implementation at **4bd8b8a**, including the new PDF-summary fallback. The major findings still apply. Review probes used synthetic temporary data; research files and saved user settings were not modified.

Proposed order. First protect identities and populations (C1, C2, C7, C10), then units (C5, C6), endpoint selection (C3), export semantics (C9) and automatic elastic fitting (C4). C8 and the additional hardening items follow or accompany the relevant stage.

Protect group and specimen identity

C1 | Stop silent sample-group pooling

Accepted: An unrelated CSV or an empty exports-named folder can turn an organisational parent into one sample group and pool its descendants. That can change group means and detach saved selections without any data moving.

Proposed remediation: Detect competing parent and descendant group boundaries before combining records. A stray file, folder name or PDF filename must not be sufficient evidence for pooling. Retain legitimate nested runs through an explicit association or a reviewed group boundary. Surface the ambiguity before producing combined results.

Acceptance checks: Sibling groups remain distinct after adding a specimen-list CSV or empty exports folder. Genuine nested instrument runs remain supported. Repeat the checks for PDF-summary discovery, arbitrary folder depth and preservation of saved group identities.

C2 | Detect duplicate exports and multiply assigned rows

Accepted: Both an original export and a re-export can load as separate specimens while claiming the same Instron row. Identical copies were independently reproduced with no check message.

Qualified remedy: Flag identical content and multiple curves assigned to one summary row. Prevent unresolved duplicates from contributing twice, but do not silently decide that the first or newest file is the correct specimen. Different curves claiming one row require an explicit resolution; files must not be deleted or silently relabelled.

Acceptance checks: Cover byte-identical copies, different data with the same row claim and legitimate distinct tests. Resolving an issue must preserve the chosen identity and its overrides on reload. Duplicate handling must also protect tested counts and location plots.

C7 | Separate a candidate match from an accepted assignment

Accepted: An Instron assignment that fails the UTS check can still supply reference statistics, gauge length and default Location. In an independent probe, the mismatched row supplied a 99 mm gauge and reconstruction was applied.

Proposed remediation: Represent assignment status explicitly. Keep candidate summary values visible for inspection, but withhold unconfirmed assignment-dependent results from Instron statistics, comparisons, reconstruction and inferred Location. A user can resolve the mapping or confirm it with a recorded reason; source changes must trigger revalidation. Keep independently valid raw-CSV results usable.

Qualification and checks: A UTS mismatch indicates uncertainty, not proof of a wrong identity. Do not use every specimen warning as an identity failure. Exercise ambiguous rows, conflicting revisions, rounding tolerance, explicit confirmation and both CSV and PDF summaries. Missing summaries remain a supported raw-only workflow.

ACCEPTED REMEDIATIONS / INPUTS

Make units and import outcomes explicit

C5 | Remove ductility-dependent strain-unit guessing

Accepted: The loader can ignore "Strain (%)" when there is no separate units row and turn a true 3% strain into 300%. This can scale members of the same group differently.

Proposed remediation: Read units from both headers and units rows; convert known fractional and percent forms explicitly. Surface conflicts. Where units cannot be established, request an explicit interpretation rather than inferring it from the strain magnitude. Retain the original channel, declared units and applied conversion in provenance.

Acceptance checks: A 3% percent-valued curve stays at 3%; a fractional value of 0.03 becomes 3%. Cover low ductility, missing units rows, contradictory declarations, unknown units and mixed groups. An uncertain interpretation must not silently become a valid result.

C6 | Separate unit conversion from channel meaning

Accepted: The current heuristics can misread kPa, GPa and psi, and can accept true stress/strain as engineering data. These are direct result-integrity issues and should be prioritised alongside C5.

Proposed remediation: Use an explicit, tested conversion table for supported stress units. Distinguish true and engineering channels independently of their units. Keep unsupported files visible with a clear reason; do not label true-only data as engineering data. An explicit true-data workflow can be designed separately.

Acceptance checks: Equivalent 475 MPa inputs expressed in each supported unit yield equivalent results. Unknown or conflicting units cannot pass silently. True-only and mixed-channel files select or reject channels predictably, with the chosen interpretation visible.

C10 | Show failed imports without redefining "tested"

Accepted: A file can appear in the specimen catalogue but fail numerical loading and disappear from the analysis with only a console message. The decimal-comma case was independently reproduced.

Qualified remedy: Do not define "tested" solely as "successfully parsed". A failed import does not undo a physical test. Show distinct counts for tests found, curves loaded and imports requiring attention; retain property-specific valid n. Keep the total planned specimen count separate. Duplicate or ambiguous records must not silently inflate the counts.

Proposed remediation: Record a per-file outcome and readable reason in the UI, including files with no loaded specimen row. Support unambiguous decimal-comma exports or reject them explicitly. Reconcile discovery and loading records so nothing disappears between these stages.

Acceptance checks: Cover semicolon-delimited decimal commas, quoted numeric commas, malformed numbers, empty or unreadable files and missing channels. Reloading a repaired file updates its status and counts without losing its identity or overrides.

Improve endpoint and yield selection

C3 | A large ratio is not sufficient evidence of a sudden drop

Accepted: A small quantised decrement following zero change can satisfy the current greater-than-five-times rule. The global noise estimate can also become zero. A later low-load reading can then confirm an earlier event that was only part of a smooth decline.

Independent evidence: A smooth synthetic tail selected the 10% crossing at 29.3075% strain when clean. Quantising stress to 0.5 MPa moved selection to 29.1200% and labelled it a confirmed sudden drop, without a review flag. This supports the mechanism; it is not a reproduction of the audit's exact numerical table. A separate noisy version of this fixture did not misclassify.

Proposed remediation: Require a candidate to be significant against the local decline, measurement noise and quantisation, with a sensible denominator floor. Validate persistence and timing as well as magnitude. Retain the progressive 10% branch and do not let an eventual low-load crossing retrospectively validate any earlier tiny decrement. Confidence wording must reflect the evidence actually present.

Qualification: A local-median rule is a candidate design, not an established replacement. Tune and validate it against abrupt, progressive and accelerating tails before adopting thresholds. Preserve acquisition order, gap handling and manual endpoint overrides.

Acceptance checks: Use clean, quantised and noisy progressive fixtures, multiple seeds and sampling densities, genuine abrupt failures, multi-reading collapses, transient load changes, acquisition gaps and truncated exports. Confirm effects on Calc EL, toughness, reconstruction and landmark endpoints, not just the detector label.

C4 | Validate the elastic region rather than assume it

Accepted: The fixed 20-50% UTS window can include plastic deformation. Independent known-modulus, known-yield curves reproduce the resulting E and YS bias. Larger failures in those probes were flagged, so this is not uniformly a silent failure.

Qualified remedy: Do not replace the fixed window with an unconstrained maximum-slope search: noise or a very short anomaly may win. Prefer candidate regions with adequate span, sufficient points, stable slope and appropriate toe/noise guards, potentially refined using a preliminary yield estimate. Retain manual fitting and show uncertainty. A high R-squared is not proof of a physically elastic region.

Policy decision: Withholding flagged YS from group results is a separate choice from improving the fitter. Define that behaviour explicitly; do not silently change populations or exclude the whole specimen. An alloy name or high YS/UTS ratio alone does not guarantee the fit is valid.

Acceptance checks: Validate against known E and Rp0.2 across YS/UTS ratios of roughly 0.3-0.9, with toe effects, sparse sampling and noise. Set numerical tolerances before tuning and compare automatic, reviewed and manual outcomes.

QUALIFIED RECOMMENDATIONS / INTERPRETATION

Keep locations and exports unambiguous

C8 | Treat inferred locations as editable interpretations

Accepted: Taking the final numeric token can turn a temperature, date or repeat number into Location. Those cases need visible provenance and a way to correct the interpretation.

Rebuttal: Multiple numeric tokens are not inherently invalid: an operator label can legitimately contain alloy grade, temperature and specimen number. Location greater than the group's specimen count is also not universally invalid when positions are numbered across an entire build. Neither rule should automatically reject data.

Proposed remediation: Show which label token supplied the default, flag plausible ambiguities and preserve explicit Location overrides. Consider a group-specific naming convention where useful. Distinguish a count of specimens from a maximum position index.

Acceptance checks: Cover alloy/temperature labels, dates, repeat tests, reserved gaps, global build positions and ordinary local numbering. Saving, resetting and reloading an override must have predictable results.

C9 | Export eligible values that can safely be aggregated

Accepted: Excluded properties can remain ordinary numeric cells in the Specimens sheet. Averaging that column can disagree with Summary even though the application's own population rules are correct.

Rebuttal: Strikethrough or coloured fill alone is insufficient: Excel AVERAGE still includes those numbers. Formatting can explain status but cannot enforce the intended population.

Proposed remediation: Use eligible/effective results in the main analysis columns, leaving ineligible entries blank. Preserve the original calculated and reported values in clearly labelled raw columns or a separate raw-data sheet, with exclusion mode and reason. Do not delete original values or rely on users to remember a filter.

Acceptance checks: For every data-use mode and global/analysis override, means, sample SDs and valid counts from the eligible export columns must reproduce the corresponding Summary values. Compare the same property and source basis, including Calc, Instron and Reconstruct. Keep full numeric precision and formula-injection protection.

Standards and property names | Clarify before claiming compliance

The audit explicitly did not verify the full ISO/ASTM clauses behind the five-times, 2% and 10% rules. This response therefore accepts the need for verification, not the authority of any unverified quotation or clause interpretation.

Verify the exact edition, clause, quantity and applicability against authorised full text before changing standards claims. Clearly distinguish total strain at a selected endpoint from permanent elongation measured after fracture. Reconstruction remains a derived model; the existence of a standards-inspired criterion does not make the entire workflow standards-compliant.

ADDITIONAL HARDENING / AUDIT SECTION 4

Accepted work and points needing evidence

These items should remain separate from confirmed defects. Where the audit did not establish a reachable user failure or an affected output, first reproduce that path before declaring the issue closed or selecting a remedy.

Item	Response and proposed action
Variable Save all	**Accept hardening.** Save edited cells and detect conflicts in those cells. The stale-draft issue was reproduced in isolation, but the audit found no reachable UI path; existing cross-session conflict checks remain relevant.
Specimen-count correction	**Accept clearer feedback.** Retain the user-requested startup/reload minimum. Show any correction and its reason, or reject a too-small manual entry clearly; avoid reporting an unchanged-looking "All variables saved".
Unsaved drafts on reload	**Accept.** Preserve compatible drafts or offer save/discard/cancel when they would be lost. Do not add a confirmation to every reload when nothing is at risk.
Two preparation paths	**Qualify.** Share source validation and provenance, and flag meaningful strain reversals. Do not force all calculations through one sorted array: fracture detection needs acquisition order, whereas plotting/interpolation may need a different view.
Pointwise SD uses ddof=0	**Verify the consumer.** Establish whether the value is shown or exported. Use sample SD where that is the stated statistic, with n=1 unavailable; otherwise label the population statistic explicitly.
Time (s) header	**Accept targeted verification.** Parse header units consistently with C5/C6 and test gap detection without a separate units row. Missing or unknown time interpretation must not silently disable a claimed safeguard.
Representative override	**Accept.** Surface ambiguous matches in the UI and require an exact stable specimen identity. Do not silently choose the first match or rely on console output.
Uniform EL at raw maximum	**Investigate before changing.** Define plateau/noise treatment and verify the relevant standard quantity. Do not silently smooth UTS identity checks when experimenting with a different uniform-elongation estimator.
Cloud-synchronised state	**Qualify.** Live behaviour was not tested. Moving locks outside synced folders may reduce stale-lock problems but does not provide cross-machine locking. Retain atomic writes, conflict checks and backups; test and document concurrent editing.
Backup retention	**Correct the premise.** Retention already includes 20 recent versions, 30 daily checkpoints and protected snapshots. Consider bounded hourly checkpoints for better intraday recovery, rather than describing time-based retention as absent.
Method-version drift	**Accept.** Align the guide's v9 wording with the implemented v10 method, then keep UI help, documentation and exported method provenance consistent.

VERIFICATION AND NEXT STEPS

Evidence, completion criteria and follow-up

Independent checks in the supported environment. These are separate synthetic probes, not claims to have rerun Claude's unavailable experiment scripts. No application code was changed for this review.

Area	Observed result
C1 / C2	Adding a stray CSV or empty exports folder collapsed two groups into their parent. Original and duplicate exports both claimed row 2 with no check failure.
C5 / C6	A percent-header curve ending at 3 became 300. Equivalent 475 MPa inputs in kPa/GPa/psi were mis-scaled; true-only columns were accepted.
C3 / C4	Quantisation triggered a false confirmed sudden-drop result (p. 4). Known E=200 GPa, Rp0.2=250 MPa curves reproduced elastic-window bias; one case returned E=59.05 GPa and YS=263.29 MPa, with a warning.
C7 / C9	A mismatched summary supplied gauge length, Location and reference statistics. An export average of 618.37 MPa differed from its eligible Summary mean of 587.45 MPa (n=2).
C8 / C10	Temperature/date/repeat labels produced misleading defaults. A decimal-comma file was recognised by the catalogue but rejected by the numerical loader.

The current supported-environment suite previously passed 406 tests. That does not cover the cases above. Claude's reported setup failures were associated with Python 3.12 rather than the required 3.13, and missing Voila/nbformat; they do not rebut the independently reproduced numerical findings.

Implementation and closure

1. Fix identity/population integrity and visible import outcomes; add explicit unit parsing next.
2. Introduce endpoint changes with adversarial controls, then align export eligibility and improve elastic-region selection.
3. Address Location and state/UI hardening without changing valid saved choices silently.
4. Close each finding only after its regression cases pass, affected UI/export behaviour is checked, and migration/provenance implications are documented. Numerical tolerances and policy choices must be specified before tuning.

Preserve the behaviour already found correct

Retain landmark curves through eligible property means, separate shape/property counts, consistent paired versus independent populations, full Excel precision, literal-string export protection, local token-protected serving, atomic/conflict-aware settings saves, verified backups and reviewed group moves. These need regression protection, not blanket replacement.

Requested follow-up from Claude

Please provide the synthetic fixtures, generators, random seeds and exact commands behind C3/C4 and the other numerical examples. Separate a confirmed user-facing failure from an isolated or code-reading concern. Please revisit the qualified remedies above, especially tested counts, Location bounds, single-path preparation, Excel formatting and existing backup retention. Supply precise standards references and the verified interpretation, identifying any point for which full-text verification remains outstanding.

Source: Claude, *Tensile Workbench - Independent Audit Report*, 26 October 2026, pp. 1-8. Audit fdcb232; checked against 4bd8b8a. Prepared for Brogan Csinger.